THE 2-FIBRE PRODUCT OF TWO STACKS IN SETOIDS OVER A STACK IN GROUPOIDS IS A STACK IN SETOIDS

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Editorial note. Special-case reduction; the cited lemma-2-fibre-product-gives-stack-in-setoids carries an 'Immediate from' proof in the source and is used as a declared reference.

Allowed standard prerequisites (editorial).

- 2-fibre product of stacks in setoids. lemma-2-fibre-product-gives-stack-in-setoids (same chapter, named). Gives the result whenever one leg is a faithful morphism from a stack in groupoids.

Source reference. Original lemma 05UI: sources/stacks-stacks/pinned-original-source.tex, source lines 1419–1419. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{C}$ be a site. Let $\mathcal{S}$ be a stack in groupoids over $\mathcal{C}$ and let $\mathcal{S}_i$, $i = 1, 2$ be stacks in setoids over $\mathcal{C}$. Let $f_i : \mathcal{S}_i \to \mathcal{S}$ be 1-morphisms. Then the 2-fibre product*

$$\mathcal{S}_1 \times_{f_1, \mathcal{S}, f_2} \mathcal{S}_2$$

is a stack in setoids over $\mathcal{C}$.

Proof. This is a special case of Lemma 05UI as f_2 is faithful. □

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